Find the Error

Grade 11 · N.CN.1

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is i^2? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Simplify $\sqrt{(-9)}$. Student's answer: ± 3

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

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Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is i^3? Student's answer: i

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

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Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Add $(2 + 3i) + (4 - i)$. Student's answer: $6 + 4i$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Multiply $(3 + i)(3 - i)$. Student's answer: 8

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Multiply $(1 + 2i)(3 + i)$. Student's answer: $5 + 7i$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $x^2 + 4 = 0$. Student's answer: $x = \pm 2$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $x^2 + 2x + 5 = 0$. Student's answer: $x = -1 \pm 2$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $x^2 - 6x + 10 = 0$. Student's answer: $x = 3 \pm 2i$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Rewrite $x^2 + 6x + 1$ by completing the square. Student writes: $(x + 3)^2 - 9$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the maximum value of $f(x) = -2x^2 + 8x$? Student's answer: It has a minimum of 8.

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the vertex of $y = x^2 - 4x + 7$? Student's answer: $(-2, 3)$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Let $p(x) = x^2 + 3x - 4$. What is $p(1)$? Student's answer: 4

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the remainder when $x^3 - 2x + 1$ is divided by $x - 2$? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Is $x - 3$ a factor of $x^2 - 5x + 6$? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $6/x = 3$. Student's answer: $x = 3$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $x/(x - 2) = 2$. Student's answer: $x = 2$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $1/x + \frac{1}{2} = 1$. Student's answer: $x = 1$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve the system $y = x^2$ and $y = x + 2$. Student's answer: (2, 4) only

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve the system $y = x^2$ and $y = 4$. Student's answer: $(2, 4)$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve the system $y = x^2 - 1$ and $y = 3$. Student's answer: $(2, 3)$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the x-values where $y = x^2$ and $y = 2x + 3$ intersect. Student's answer: $x = 3$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find where y = 2^x and y = x + 3 intersect. Student says: They intersect at x = 3.

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the x-values where $y = x^3$ and $y = x$ intersect. Student's answer: $x = 1$ only

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. Student's answer: 8

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 11 · F.IF.6

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the average rate of change of $f(x) = 3x - 2$ on $[0, 4]$. Student's answer: 12

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the average rate of change of $f(x) = 2^x$ on $[0, 2]$. Student's answer: 3

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Where is the vertex of $f(x) = |x - 2| + 1$? Student's answer: $(-2, 1)$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the domain of $f(x) = \sqrt{(x + 4)}$? Student's answer: $x \geq 4$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Let $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. What is $f(-2)$? Student's answer: 4

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Is 8^x equivalent to 2^(3x)? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A population is modeled by y = 1000(1.05)^t. What is the annual growth rate? Student's answer: 1.05%

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Rewrite 3^(2t) in the form b^t. Student's answer: 6^t

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$f(x) = 2x + 5$ and g has a constant rate of change of 3. Which function has the greater rate of change? Student's answer: f

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Which grows faster eventually: $f(x) = 2^x$ or $g(x) = x^2$? Student's answer: $g(x) = x^2$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$y = 4x + 1$ and $y = x^2 + 3$ are compared. Which has the greater y-intercept? Student's answer: $y = 4x + 1$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Let $f(x) = x^2$ and $g(x) = f(x) + 3$. What is $g(2)$? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Let $g(x) = f(x - 2)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Let $g(x) = 2f(x)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the inverse of $f(x) = 2x + 3$. Student writes: $f^{-1}(x) = (x + 3)/2$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

If $f(x) = x^3$, what is $f^{-1}(8)$? Student's answer: 4

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Is $f(x) = x^2$ one-to-one over all real numbers? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is $\log_2 8$? Student's answer: 4

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is $\log_{10} 1000$? Student's answer: 2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Solve $2^x = 32$. Student's answer: $x = 4$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Convert 180° to radians. Student's answer: 2π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Convert 90° to radians. Student's answer: π/3

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An arc has radius 5 and central angle 2 radians. What is the arc length? Student's answer: 7

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is $\cos(\pi/2)$? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is sin π? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the point on the unit circle at 60°? Student's answer: $\left(\sqrt{\frac{3}{2}}, \frac{1}{2}\right)$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the amplitude of $y = 3\sin(2x)$? Student's answer: 2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the period of $y = \sin(2x)$? Student's answer: 2π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the period of $y = \cos(x/3)$? Student's answer: 3π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 11 · S.ID.6

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A model is $y = 2x + 1$. What is the predicted y-value when $x = 5$? Student's answer: 10

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An actual value is 12 and the predicted value is 11. What is the residual? Student's answer: -1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A scatter plot has a correlation coefficient of r = 0.95. Is the linear relationship weak or strong?

Student's answer: weak positive

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 11 · S.ID.7

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A taxi cost is C = 25 + 4m, where m is minutes. What does the slope 4 mean? Student's answer: $25 per minute

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 11 · S.ID.7

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In the same taxi model C = 25 + 4m, what does the intercept 25 mean? Student's answer: the cost per minute

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 11 · S.ID.7

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A temperature model is T = 80 - 2h, where h is hours. What does the slope mean? Student's answer: The temperature rises 2 degrees per hour.

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.